

Contrastive Attribution for Teacher-Student Language Models

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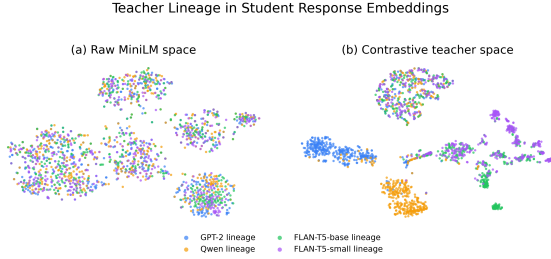


Figure 1: t-SNE visualization of student response embeddings before and after contrastive training. Colors indicate documented teacher lineage.

Abstract

Teacher attribution asks whether the outputs of a student language model reveal which teacher model shaped it. Prior work mainly detects teacher traces with lexical and syntactic features, but these signals can mix teacher style with prompt content. This project tests whether a modern contrastive representation can improve teacher detection for public teacher-student lineages. For each prompt, candidate teachers and students generate responses; a prompt-conditioned MiniLM encoder is trained to pull each student response toward its documented teacher and away from same-prompt alternatives. On a balanced four-way public-lineage benchmark, direct contrastive retrieval reaches 60.44% single-prompt accuracy, while a frozen logistic-regression probe over the learned embedding reaches 63.12% accuracy and 89.11% macro ROC-AUC, slightly above the strongest replicated lexical baseline at 62.17%. Aggregating multiple prompts strengthens the lineage signal, reaching 97.5% accuracy with eight prompts and 100% with sixteen. The results suggest that teacher fingerprints survive public distillation, but that reliable attribution depends on task, aggregation, and stronger representations.

1 Introduction

Knowledge distillation transfers capabilities from a large teacher language model into a smaller student (Xu et al., 2024). It improves efficiency, but black-box distillation also creates a provenance problem: a student can be trained from teacher responses without access to the teacher’s weights or data (Chen et al., 2024). Recent public debates about whether open-source models distill from stronger closed-source systems motivated the question of this project: **can modern representation learning substantially improve teacher detection from model outputs?** Reliable teacher attribution would help audit public model lineages and reveal which behavioral traces survive distillation, giving a better view of what is actually transferred between language models.

Li et al. (2024) study origin attribution for LLM-generated text, while Wadhwa et al. (2025) focus on teacher tracing in controlled distillation. This project adapts the question to a messier setting: public teacher-student model pairs whose model cards document a teacher or source model.

Public lineages are realistic but noisy: model cards vary, students may inherit from a family rather than one clean teacher, and checkpoints may include additional instruction tuning. To reduce topic leakage, all candidate teachers and students answer the same prompts, producing examples with one student response, several same-prompt teacher responses, and a documented teacher label.

The modeling idea comes from contrastive authorship attribution (Ai et al., 2022; Hu et al., 2024; Huertas-Tato et al., 2024): if authors leave stylistic traces separable from topic, teacher models may leave behavioral traces in their students. The proposed model therefore contrasts each student response against same-prompt teacher responses.

The main contributions are:

- a reproducible public-lineage teacher-

Teacher ID	Teacher	Student
gpt2	gpt2	distilgpt2
qwen15_18b	Qwen1.5-1.8B	MiniPLM-Qwen-200M
flan_t5_base	FLAN-T5-base	LaMini-FLAN-T5-248M
flan_t5_small	FLAN-T5-small	LaMini-FLAN-T5-77M

Table 1: Public teacher-student pairs used for attribution.

attribution pipeline;

- a prompt-conditioned contrastive encoder for same-prompt student-teacher comparison;
- an evaluation showing when the learned representation improves over lexical baselines and when task structure still dominates.

2 Experiments

2.1 Teacher-Student Pairs

The main experiment uses four public teacher-student lineages (Table 1), giving a balanced four-way classification task with 25% chance accuracy. Two pairs have explicit model-card evidence; the FLAN-T5 pairs are medium-confidence same-family instruction-distilled lineages.

2.2 Prompt Bank and Data

The prompt bank follows Wadhwa et al. (2025): summarization, question answering, and instruction following. It draws from CNN/DailyMail, SumpubMed, Rotten Tomatoes, CommonsenseQA, OpenBookQA, QuaRel, and Alpaca-style instructions. All teachers and students answer the same prompts. After filtering, the run contains 29,742 training prompts and 3,000 test prompts. Each prompt contributes one row per teacher label, so the test split has 12,000 balanced attribution rows with prompt, student response, candidate teacher responses, true label, task, and dataset metadata.

2.3 Contrastive Attribution Model

Each input is a prompt-response pair (x, y) encoded by a shared sentence-transformers/all-MiniLM-L6-v2 encoder with a projection head and teacher-classification head. Pairs are tokenized with the MiniLM tokenizer and truncated to 512 tokens. The 29,742 training prompts yield 118,968 attribution rows; with one student and four teacher encodings per row, this is at most about 305M MiniLM token positions per epoch before padding removal.

For each row, the student embedding is the anchor, the documented teacher is positive, and the other same-prompt teachers are negatives:

$$\mathcal{L}_{\text{NCE}} = -\log \frac{\exp(\text{sim}(z_s, z_{t^+})/\tau)}{\sum_k \exp(\text{sim}(z_s, z_{t_k})/\tau)}, \quad (1)$$

where sim is cosine similarity. The final objective adds supervised classification:

$$\mathcal{L} = \mathcal{L}_{\text{NCE}} + \lambda \mathcal{L}_{\text{CE}}. \quad (2)$$

The final run uses $\tau = 0.05$, $\lambda = 0.2$, a 128-dimensional projection, and validation-accuracy early stopping.

At evaluation time, the learned representation is used in three ways. First, direct cosine retrieval chooses the candidate teacher response closest to the student response in the contrastive space. Second, the classifier head predicts the teacher ID directly from the student embedding. Third, an embedding probe freezes the encoder and trains logistic regression on student embeddings from the training split. This probe tests whether teacher identity is linearly recoverable from the learned representation, even when raw cosine retrieval underuses it.

2.4 Baselines and Metrics

The experiments reproduce *Who Taught You That?* (WTYT) baselines (Wadhwa et al., 2025): response-only BoW and TF-IDF similarity, BoW logistic regression, and 1–4 gram logistic regression. POS templates are omitted from the replicated table because their template-mining procedure was not reproduced reliably; their published POS results are used only as an external reference.

The evaluation reports top-1 accuracy, top-2 accuracy, one-vs-rest ROC-AUC, confusion matrices, task and dataset breakdowns, and set-level accuracy. Set-level attribution aggregates evidence from multiple responses from the same student lineage before predicting the teacher.

3 Results

3.1 Single-Prompt Attribution

Table 2 shows the current main contrastive result and WTYT-style baselines. Direct contrastive retrieval reaches 60.44% accuracy and 82.84% top-2 accuracy on 12,000 test rows, far above 25% chance. The frozen embedding probe reaches 63.12% accuracy and 89.11% macro ROC-AUC,

Method	Acc.	Top-2	AUC
BoW response cosine	32.83	58.63	54.82
TF-IDF response cosine	34.23	60.10	55.33
BoW classifier	58.36	87.58	81.89
1–4 gram classifier	62.17	88.14	83.89
Contrastive cosine retrieval	60.44	82.84	68.15
Contrastive classifier head	62.90	83.29	88.39
Contrastive embedding probe	63.12	83.84	89.11

Table 2: Test results. The embedding probe is logistic regression on frozen contrastive student embeddings.

slightly above the strongest replicated lexical baseline, the 1–4 gram classifier at 62.17%. This suggests that the learned embedding contains teacher information not fully exploited by raw cosine matching.

Performance varies by task: the embedding probe reaches 84.63% on QA, 68.20% on instruction following, and 47.08% on summarization. Dataset results match this pattern: Rotten Tomatoes, CommonsenseQA, and OpenBookQA exceed 83%, while CNN/DailyMail and SumpubMed remain near chance. QA is the easiest setting, likely because answers preserve formatting, reasoning habits, and label choices; summarization is hardest because compression pushes different models toward similar high-level summaries.

3.2 What the Representation Learns

Figure 1 gives a qualitative view of the learned space. In raw MiniLM embeddings, points of different teacher lineages are heavily mixed; the visible clusters are better explained by shared prompt type or dataset than by teacher identity. After contrastive training, GPT-2 and Qwen responses form more coherent lineage regions, and FLAN-T5-base and FLAN-T5-small become more separated than in the raw space. The figure is still not perfectly clustered by lineage. The large mixed region at the top of the contrastive panel is likely a residual task or dataset region rather than a teacher region: the quantitative breakdown shows that summarization is the weakest task, and summarization datasets are close to chance. This supports the main interpretation of the project: contrastive learning increases the visibility of teacher fingerprints, but it does not fully remove content and task effects.

3.3 Comparison to WTYT

The comparison to Wadhwa et al. (2025) is not one-to-one: they train controlled GPT-2 and OLMo students from five teachers, while this project uses

four public lineages. In their Table 2, POS templates average about 58.2% accuracy, versus 53.9% for 1–4 grams and 44.2% for their BERT-based model, a 7.9% relative gain over n-grams. On the public-lineage data, the embedding probe, a logistic-regression classifier on frozen contrastive student embeddings, improves over the replicated n-gram classifier by 0.95 points, or 1.5% relative. It is also above their average POS result (63.12% vs. about 58.2%), but this is only a cross-experiment reference, not evidence of beating POS templates on an identical benchmark.

3.4 Set-Level Attribution and Error Analysis

Attribution improves sharply when prompts are aggregated (Figure 2). Here, aggregation means sampling several prompt-response pairs from the same student lineage and averaging their teacher scores before making one teacher prediction. For contrastive cosine retrieval, accuracy rises to 73.75% with two prompts, 90.0% with four, 97.5% with eight, and 100% with sixteen. The confusion matrices show why: single-prompt predictions often collapse Qwen and FLAN outputs toward GPT-2, while aggregation removes most off-diagonal mass. By four prompts, the main remaining ambiguity is between the two FLAN lineages; by sixteen prompts, all sampled sets are classified correctly. This matches the single-prompt error analysis: lineages that are close in model family require more evidence, while lineages with more distinctive response behavior separate quickly.

3.5 Ablations and Training Behavior

Seven fixed five-epoch MiniLM ablations vary temperature, classification weight, learning rate, and projection dimension. The best test ablation was 1e5 at 60.08%, below the full early-stopped MiniLM model; the best validation ablation reached 59.68% on test. Together with the training curve, this suggests that MiniLM is close to saturated under this setup: loss keeps falling while validation accuracy flattens around 60–61%. Future work should test larger or more style-sensitive encoders. Significance tests were not run, so small margins such as the 0.95 point gain over n-grams should be interpreted cautiously.

4 Related Work

Knowledge distillation compresses or specializes LLMs (Xu et al., 2024), and black-box distillation shows that this can happen through gener-

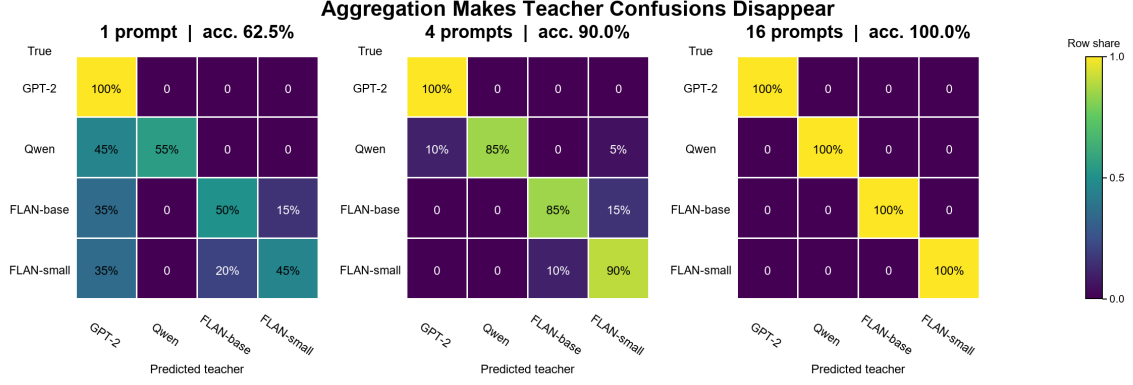


Figure 2: Row-normalized confusion matrices for set-level contrastive cosine retrieval. Rows are true teachers and columns are predicted teachers. Aggregating multiple responses from the same student lineage makes the teacher fingerprint more stable and moves mass toward the diagonal.

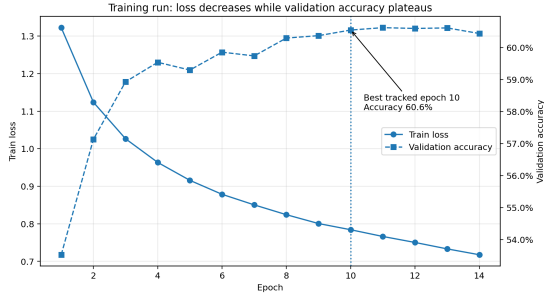


Figure 3: Training loss continues to decrease after validation accuracy begins to plateau, suggesting that the compact MiniLM encoder is close to saturated under this setup.

ated outputs alone (Chen et al., 2024). This motivates output-based provenance. Li et al. (2024) study source-model attribution for LLM text, while Wadhwa et al. (2025) introduce teacher tracing in controlled distillation and show that syntactic templates can preserve teacher footprints. This work keeps the tracing question but uses documented public lineages and a contrastive same-prompt encoder.

The contrastive component follows InfoNCE and supervised contrastive learning (van den Oord et al., 2018; Khosla et al., 2020), and is motivated by authorship attribution. Ai et al. (2022), Hu et al. (2024), and Huertas-Tato et al. (2024) show that contrastive encoders can learn source-specific representations while reducing topic dependence. Teacher attribution has the same goal: capture response style and behavior rather than prompt topic.

5 Conclusion

This project builds a public-lineage teacher attribution pipeline to test whether modern representation learning improves teacher detection. On a four-way task, MiniLM reaches 60.44% with direct retrieval and 63.12% with a frozen linear probe, slightly above the strongest replicated lexical baseline and competitive with the average POS-template result of Wadhwa et al. (2025). The improvement is not large enough to claim a decisive single-prompt breakthrough, but the latent-space visualization and set-level confusion matrices show that contrastive training makes teacher lineage more recoverable. Set-level aggregation performs much better, supporting the idea that student outputs preserve teacher fingerprints across prompts. Future work should add more verified lineages, test larger style-sensitive encoders, and evaluate harder cross-dataset or paraphrased-prompt settings.

Limitations

This project uses public model-card lineages rather than controlled distillation, so labels are realistic but noisier than in a laboratory setup. The four-way label space is small, and two classes are closely related FLAN-T5-family models. The same-prompt setting assumes candidate teacher responses are available for the prompts being evaluated. Generation settings may also affect attribution because decoding style can change surface behavior. The POS-template method from Wadhwa et al. (2025) is compared only through their published results, not rerun on the same public-lineage data. Finally, the encoder experiments use a compact MiniLM backbone; larger or more authorship/style-sensitive

encoders may yield stronger attribution results.

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References

- Bo Ai, Yuchen Wang, Yugin Tan, and Samson Tan. 2022. [Whodunit? learning to contrast for authorship attribution](#). In *Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 1142–1157, Online only. Association for Computational Linguistics.
- Hongzhan Chen, Ruijun Chen, Yuqi Yi, and 1 others. 2024. [Knowledge distillation of black-box large language models](#). *arXiv preprint arXiv:2401.07013*.
- Zhiqiang Hu, Thao Thanh Nguyen, Yujia Hu, and 1 others. 2024. [Contrastive disentanglement for authorship attribution](#). In *Companion Proceedings of the ACM Web Conference 2024*, pages 1657–1666.
- Javier Huertas-Tato, Adrian Giron-Jimenez, Alejandro Martin, and David Camacho. 2024. [Isolating authorship from content with semantic embeddings and contrastive learning](#). *arXiv preprint arXiv:2411.18472*.
- Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschiot, Ce Liu, and Dilip Krishnan. 2020. Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, volume 33, pages 18661–18673.
- Liyang Li, Yihan Bai, and Minhao Cheng. 2024. [Where am i from? identifying origin of LLM-generated content](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.
- Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. [Representation learning with contrastive predictive coding](#). *arXiv preprint arXiv:1807.03748*.
- Somin Wadhwa, Chantal Shaib, Silvio Amir, and Byron C. Wallace. 2025. [Who taught you that? tracing teachers in model distillation](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 3307–3315, Vienna, Austria. Association for Computational Linguistics.
- Xiaohan Xu, Ming Li, Chongyang Tao, and 1 others. 2024. [A survey on knowledge distillation of large language models](#). *arXiv preprint arXiv:2402.13116*.